GLOBAL LITHIUM CARBONATE MARKET

Mine-Level Supply Analysis and 2026–2029 Trading Window Framework

Supplement to: Global Lithium Carbonate Market Analysis 2025–2030

Xueting Xiang | PhD Researcher, Quantitative Finance | Universiti Malaya | May 2026
xiangxuetingum@gmail.com | insightgatenews.com

Research Assistance: This report was produced with analytical support from Claude (Anthropic) and GPT-4 (OpenAI), used for data structuring, scenario framework development, and document drafting. All analytical judgments, source selection, and conclusions are the author's own.

Data Sources: All production data and market information in this report are sourced exclusively from publicly available materials, including company annual and quarterly reports, ASX announcements, earnings releases, and publicly accessible industry media (SMM, Mysteel, Reuters, Bloomberg). This analysis does not draw on proprietary data from commercial databases such as Fastmarkets, Wood Mackenzie, Benchmark Mineral Intelligence, or S&P Global Commodity Insights, to which the author does not have licensed access. Where proprietary data would materially improve accuracy, this is noted explicitly in the text.

Executive Summary

The global lithium carbonate market has recovered from its June 2025 floor of USD 8,259/t to approximately USD 24,000/t in April 2026. This report examines whether that recovery is supported by mine-level supply fundamentals — and what the evidence implies for procurement and trading strategy through 2029.

Three Core Findings

- The recovery is structurally supported. Three simultaneous supply shocks — the Zimbabwe concentrate export ban (February 2026), CATL's Jiangxi lepidolite mine suspension (August 2025), and Greenbushes production guidance downgrade — have removed material marginal supply at exactly the moment the market was transitioning from surplus to deficit.
- The 2027–2028 window is the highest-conviction period. The investment drought of 2023–2024 removed 30–40% of the planned supply pipeline. Mine development takes 5–10 years. The missing supply of 2027–2028 cannot be recovered regardless of current investment decisions. This is not a forecast — it is a mechanical consequence of development timelines.
- Sodium-ion substitution creates a structural ceiling at approximately USD 18,000–20,000/t for grid storage applications. A repeat of the 2022 supercycle (USD 80,000/t) is structurally impossible. The recovery corridor is bounded: USD 15,000–25,000/t is the analytically defensible range.

Procurement Recommendation Summary

Scenario	Supply Direction	Recommended Action
Base Supply	Stable to moderate growth; Arcadia	Normal procurement cadence; moderate

Scenario	Supply Direction	Recommended Action
	sulphate transition buffers Zimbabwe ban	spot exposure acceptable
Tight Supply	Marginal supply contraction; lepidolite + non-sulphate Zimbabwe exit	Increase long-term offtake exposure; secure Australian SC6 supply NOW
Loose Supply	Optional supply re-entry: Mt Cattlin, Manono, Jiangxi recovery	Hold spot exposure; defer long-term commitments; wait for CAAM confirmation

Current evidence (April 2026 prices at ~USD 24,000/t; GFEX inventory below 40,000 tonnes; CAAM March NEV sales +18% YoY) is most consistent with the Tight Supply case materialising faster than the Base case timeline. This creates a time-sensitive procurement window.

Part I — Why Mine-Level Analysis?

1.1 The Limitation of Aggregate Supply Forecasts

Most lithium carbonate market reports present supply forecasts at the regional or country level: 'Australia will produce X million tonnes of spodumene concentrate in 2026.' These aggregate figures conceal the heterogeneity that actually determines price behaviour.

Consider what a country-level figure hides:

- Greenbushes (cash cost ~USD 5,500/t LCE equivalent) and Wodgina (cash cost ~USD 9,500/t) are both 'Australian hard rock supply' — but they respond very differently to a price decline below USD 10,000/t.
- Mt Cattlin entered care and maintenance in March 2025. It is still counted in many aggregate 'Australian capacity' figures as a producing asset.
- Jiangxi lepidolite and Sichuan hard rock are both 'Chinese supply' — but lepidolite costs USD 11,000–14,000/t to produce, while Jiajika (Sichuan) hard rock benefits from cost advantages. They have opposite positions on the cost curve.

The critical analytical question is not 'how much supply exists globally' but 'which supply will exit at current prices, and how quickly can the remainder respond to a price recovery?' That question requires mine-level data.

1.2 What Mine-Level Data Can Answer

- Price floor identification: Which mines are producing below cash cost at current prices? These are the candidates for closure, and their exit is the self-correcting mechanism that ends every commodity downturn.
- Supply elasticity: How long does it take for mothballed capacity to restart? For Australian hard rock, the answer is 3–6 months for a care-and-maintenance restart, and 5–7 years for a greenfield development.
- Procurement counterparty risk: Is Huayou's Arcadia project insulated from Zimbabwe's concentrate export ban? The answer requires knowing that Huayou commissioned a lithium sulphate plant in October 2025 — data that does not appear in aggregate supply tables.
- Cost curve precision: The general market knows the hard rock cash cost floor is approximately USD 8,000–9,000/t. Mine-level data allows us to identify which specific operations set that floor and at what volume.

1.3 Data Availability and Methodology

Mine-level lithium data availability varies dramatically by asset type and geography. This analysis uses a three-tier source hierarchy:

Grade	Source Type	Examples	Typical Assets
A	Company annual/quarterly report	IGO FY25, Pilbara Minerals FY24, Lithium Argentina 2025 Production	Greenbushes, Pilgangoora, Cauchari-Olaroz
B	Industry data provider	SMM, Mysteel, Benchmark Mineral Intelligence	Jiangxi lepidolite (regional), Bikita
C	News media / secondary	Reuters, Bloomberg	Zimbabwe policy, Manono status

LCE (lithium carbonate equivalent) conversion uses a market-convention factor of ×0.22 for SC6-equivalent spodumene concentrate. This is a supply-side normalisation tool, not a chemically precise conversion. Where mine-level product is SC5.5 or non-standard, this approximation introduces a margin of error of approximately 8–10%. All figures in the database are clearly marked with their original units and conversion methodology.

Critical data limitation: Wodgina's 100% mine basis production requires quarterly reconstruction because MinRes's ownership interest increased from 40% to 50% on 18 October 2023. The reconstruction methodology — using company-disclosed 100% basis figures for Q2 FY24 (which straddles the ownership change) and applying the appropriate divisor to other quarters — is documented in full in the companion Excel database (Sheet 2: Sources). This is a known complexity in Wodgina analysis and is not a unique limitation of this dataset.

Part II — Data: Reading the Mine Supply Database

2.1 How to Use the Companion Excel Database

The companion Excel file contains three worksheets that together form the evidentiary foundation for this report's conclusions. They should be read alongside this document.

Sheet 1: Mine Data (LCE)

The primary data table lists 13 mine assets and regional supply pools across five geographies. Each row contains:

- Original production figures in the company's native units (spodumene concentrate kt, lithium carbonate tonnes, regional LCE aggregate)
- LCE-converted figures for 2022–2025, using the ×0.22 SC6 convention for hard rock and ×1.00 for carbonate/hydroxide
- A confidence grade (High / Med-High / Med / Low) reflecting the source quality
- Current operating status, which is critical for supply modelling

The colour coding in the LCE columns uses green for positive confirmed figures and grey for estimated or pending data. Empty cells represent genuinely missing data — not zero production. The distinction matters.

Sheet 2: Sources and Original Text

Every data point in Sheet 1 has a corresponding row in Sheet 2 with the source grade, originating document, verbatim text excerpt or paraphrase, and URL or reference. This allows the reader to verify any figure independently and understand exactly what each number represents.

Source grade A entries can be relied upon for financial decisions. Grade B entries are suitable for directional analysis but should be cross-checked before use in contracts. Grade C entries provide context and should be treated as indicative only.

Sheet 3: Supply Scenarios 2026–2029

The scenario table is a directional framework, not a volume forecast. It organises mine assets into three supply cases (Base, Tight, Loose) based on which assets are assumed to be active, and provides year-by-year directional judgments alongside procurement action recommendations. The scenario table is designed to be updated as new quarterly data arrives — it is a living framework, not a static prediction.

2.2 Australia — Hard Rock (Spodumene)

Australian hard rock mines account for the majority of global spodumene concentrate supply and provide the clearest production data due to ASX disclosure requirements. Four operations dominate.

Greenbushes — The Market Anchor

Greenbushes is the world's largest and highest-grade hard rock lithium mine. Operated by Talison (51% Tianqi, 49% Albemarle via IGO), it functions as the global cost anchor: when prices fall below Greenbushes' cash cost, the entire marginal cost structure of the market is compromised.

Metric	FY22	FY23	FY24	FY25
Production (kt SC6)	1,135	1,491	1,383	1,479
LCE equivalent (kt)	249.7	328.0	304.3	325.4
Cash cost (A$/t sold)	238	279	330	325
Source	IGO FY22 AR	IGO FY23 AR	IGO FY24 AR	IGO FY25 Results

Two aspects of the Greenbushes data warrant particular attention. First, FY24 production fell 7% from FY23 despite prices that were still well above marginal cost — indicating that ore grade and weather-related operational factors create production variability independent of price signals. Second, FY26 guidance has been downgraded from the original 1,500–1,650kt to 1,375–1,425kt (per Benchmark Mineral Intelligence), a reduction of approximately 125–225kt — translating to a loss of 27–50kt LCE from the world's most important single supply source.

Pilgangoora — The Cost-Reduction Story

Operated 100% by Pilbara Minerals, Pilgangoora has consistently reduced its FOB unit operating cost from A$613/t in FY23 to a guidance of A$560–600/t in FY26 — the most significant cost improvement of any major hard rock operation. FY26 production guidance of 820–870kt represents the single largest year-on-year volume increase among currently operating Australian mines, adding approximately 15–25kt LCE to global supply versus FY25.

The Pilgangoora BMX (Battery Material Exchange) auction platform provides a real-time spodumene price signal. BMX auction prices function as the input cost reference for Chinese refiners — the equivalent of crude oil in the refining analogy. When BMX prices rise faster than GFEX carbonate prices, refinery margins compress and production cuts follow within weeks.

Mt Cattlin — Care and Maintenance, Not Zero

Mt Cattlin (now owned by Rio Tinto following the Arcadium acquisition) entered care and maintenance in March 2025 following sustained prices below production cost. Its historical production of approximately 130kt SC6 / 29kt LCE per year should not be written off entirely — Rio Tinto has indicated the possibility of restart and underground extension studies — but it cannot be included in Base or Tight supply case modelling. Any restart decision requires a sustained price recovery above approximately USD 18,000–20,000/t for at least two to three quarters.

Wodgina — The Complexity Case

Wodgina's production data requires the most careful handling in this dataset. MinRes's ownership interest increased from 40% to 50% on 18 October 2023 following the MARBL JV restructure. This means that MinRes's published attributable production figures for FY24 cannot be converted to a 100% mine basis using a single divisor.

The reconstruction methodology used in the Excel database (Sheet 2, Wodgina row) applies: Q1

FY24 attributable ÷ 40%; Q2 FY24 uses the company-disclosed 100% basis of 115kt directly (because the restructure occurred mid-quarter); Q3–Q4 FY24 attributable ÷ 50%. This produces an estimated FY24 100% basis production of approximately 451kt SC6 / 99kt LCE, labelled 'Med-High' confidence because of the reconstruction's dependence on quarterly data.

2.3 South America — Brine Operations

Salar de Atacama — The Largest Single Source

The Salar de Atacama hosts two distinct operations: SQM and Albemarle. Their production data is published at the company level, not the mine level — a critical distinction when using these figures for supply modelling.

SQM's 2024 lithium sales of approximately 205kt LCE include a small contribution (approximately 4kt) from the Mt Holland hard rock project in Australia. Albemarle does not publish a standalone Salar de Atacama production figure in its public disclosures. Therefore, the Atacama figures in the database represent SQM company-level lithium sales (Grade A, from SQM earnings releases) and an Albemarle entry marked as data not available — not an absence of production, but an absence of accessible public data at the required level of specificity.

The SQM-Codelco JV, which received all required regulatory approvals by December 2025 (Chile FNE, Chinese SAMR, Chilean Comptroller), introduces governance uncertainty into the largest single source of global lithium carbonate supply. The JV's impact on production volumes is not yet quantifiable, but the governance transition creates a risk factor that aggregate supply models systematically ignore.

Cauchari-Olaroz — The New Entrant

Cauchari-Olaroz (Lithium Argentina + Ganfeng, 40,000 tpa Stage 1 nameplate) represents the most significant new brine supply source to reach commercial production in the 2023–2025 period. The 2025 production of 34,100t lithium carbonate at a Q4 2025 cash operating cost of USD 5,618/t places it firmly in the lower-cost half of the global supply curve. Its 2026 guidance of 35,000–40,000t suggests it is approaching nameplate capacity — incremental upside beyond 40kt LCE per year would require the Stage 2 expansion decision.

2.4 China — The Marginal Cost Battleground

Jiangxi / Yichun Lepidolite — The Price-Sensitive Swing Producer

Jiangxi lepidolite is not a single mine. It is a regional supply pool comprising dozens of operations in the Yichun area of Jiangxi Province, operated by CATL, Yongxing Materials, Jiangxi Special Electric Motor, and others. The impossibility of treating it as a single mine is not a data limitation — it reflects the actual structure of this supply source.

SMM and securities firms research report regional lepidolite LCE production at approximately 104kt in 2022, 128kt in 2023. The 2024 figure is contested across sources — estimates range from approximately 75kt LCE (Guotai-Haitong) to undefined higher figures — reflecting genuine operational volatility and measurement challenges. For 2025, SMM's estimate of over 180kt LCE

reflects the ramp-up of new capacity at prevailing prices, but this figure must be read in the context of CATL's Jiangxi lepidolite mine suspending operations in August 2025 when its mining permit expired, pending renewal.

Analytical implication: Jiangxi lepidolite should be modelled as a regional marginal supply pool with a cost range of approximately USD 11,000–14,000/t LCE equivalent. At prices below this range, the pool contracts. At prices above this range, it expands. The uncertainty in 2024 production data does not reduce the analytical usefulness of this framework — it reinforces it. The data uncertainty itself reflects the price-sensitive, fragmented nature of this supply source.

Jiajika 134 (Sichuan) — The Hard Rock Alternative

Jiajika, operated by Rongjie Shares / Rongda Lithium, represents China's most significant domestic hard rock lithium resource. Its 2025 production of 185,600 tonnes of 6% grade lithium ore (spodumene), representing a year-on-year increase of 175%, is the most striking production growth figure in the global dataset for 2025. However, this growth must be understood correctly: it results from outsourced processing (external outsourced mineral processing), not from expansion of the mine's own processing capacity. The underlying resource remains under-developed relative to its potential — a distinction that matters for medium-term supply projections.

2.5 Africa — The Geopolitical Variable

A Critical Distinction: Bikita vs. Arcadia

This is the most important factual correction in this dataset. Bikita is a Sinomine (中矿资源) asset, NOT a Huayou Cobalt asset. The confusion arises because both companies operate in Zimbabwe and both are Chinese-owned. The Huayou Cobalt (华友钴业) asset in Zimbabwe is Arcadia / Prospect Lithium Zimbabwe. These are two entirely separate projects with different ownership, different geological profiles, and — critically for this analysis — different exposure to Zimbabwe's export policy.

Arcadia / Prospect Lithium Zimbabwe (Huayou Cobalt)

Huayou Cobalt's 2024 Annual Report confirms that Arcadia shipped approximately 400,000 tonnes of lithium concentrate in 2024, representing a year-on-year increase of approximately 41% (implying 2023 shipments of approximately 284kt). This places Arcadia at approximately 88kt LCE equivalent in 2024 — a meaningful contribution to global supply.

The strategically significant development for 2025–2026 is the commissioning of Huayou's 50,000 tpa lithium sulphate plant in October 2025 and the first lithium sulphate export from Zimbabwe in April 2026. Zimbabwe's February 2026 export ban explicitly covers lithium concentrate — it does not cover lithium sulphate or other processed lithium compounds. Huayou's early investment in local processing capability therefore insulates Arcadia from the policy shock that disrupts competitors who export only raw concentrate.

Bikita (Sinomine / 中矿资源) — As a Competitive Supply Signal

For analysis purposes related to Huayou Cobalt, Bikita should appear in the competitive supply context, not the proprietary asset column. Bikita's 2023 production of approximately 250kt

concentrate (55kt LCE equivalent) and its exposure to Zimbabwe's export restrictions (not offset by equivalent local processing capability) mean that it faces greater disruption from the 2026 policy environment than Arcadia. This is relevant context for understanding Zimbabwe's aggregate supply impact on global markets.

Manono (DRC) — Optionality, Not Forecast

Manono should appear in the supply database at zero commercial production for all years through 2025. The AVZ Minerals arbitration against the DRC government and various counterparties remains unresolved. Reuters reported in February 2026 that Zijin Mining and Cominière plan to initiate DRC's first lithium output from the Manono area in June 2026 — but this is a media report without confirmed production volumes, and the legal ownership situation remains contested. The AVZ DFS capacity of 700,000 tpa SC6 is a geological resource estimate, not a supply forecast, and must not be used as such.

Part III — Key Findings

3.1 Finding One: The Investment Drought Is Quantifiable

The generic market observation that 'investment collapsed during the 2023–2024 price downturn' can now be grounded in specific asset decisions visible in the mine-level data:

- Mt Cattlin (29kt LCE capacity): Care and maintenance from March 2025. No restart timeline given. Conditional on price recovery above approximately USD 18,000–20,000/t.
- Greenbushes FY26 guidance downgrade: 125–225kt SC6 reduction / 27–50kt LCE reduction from a single guidance revision at the world's largest hard rock mine.
- CATL Jiangxi lepidolite (approximately 46kt LCE annual capacity): Suspended August 2025 on mining permit non-renewal. Restart conditional on permit approval.
- Albemarle Wodgina: Albemarle deferred expansion decisions; all three processing trains remain available but are not running at nameplate capacity.
- Manono: The world's largest undeveloped lithium resource remains in legal dispute. No development timeline is credible.

The combined effect of these specific, documentable decisions — not aggregate statistics — is the 30–40% supply pipeline reduction figure cited in the primary lithium carbonate report. Here, for the first time, that estimate has names and addresses.

3.2 Finding Two: The 2026 Price Level Is Supported by the Cost Curve

At approximately USD 24,000/t (April 2026), the current price sits above the cash cost of every operating mine in the database. This means the market is not being price-supported by low-cost production alone — it reflects genuine supply tightness. But more importantly, it means the recovery has room to sustain.

The cost curve, constructed from mine-level data, reveals the following structure:

Mine / Region	Cash Cost (approx. USD/t LCE)	Status	Price Signal
Greenbushes	~5,500–6,500	Operating	Always profitable above USD 8,000
Cauchari-Olaroz	~5,600 (Q4 2025)	Operating	Confirmed below floor
Pilgangoora	~8,000–9,500 (FOB equiv.)	Operating	Marginally above June 2025 floor
Wodgina	~9,500–11,000	Operating	Profitable at current prices
Jiangxi Lepidolite	~11,000–14,000	Volatile	At/below cost when price <USD 14,000
Mt Cattlin	~10,000–12,000	Care & maint.	Restart requires >USD

Mine / Region	Cash Cost (approx. USD/t LCE)	Status	Price Signal
			18,000–20,000

The June 2025 floor of USD 8,259/t is not coincidental. It corresponds precisely to the cash cost of hard rock production in the Pilgangoora range — the self-correcting mechanism that ends every commodity downturn. Below this price, operators reduce or halt production. The market's supply response mechanism worked exactly as the cost curve framework predicts.

3.3 Finding Three: Zimbabwe Policy Creates Asymmetric Risk by Operator

Zimbabwe's February 2026 export ban on lithium concentrate is not a uniform supply shock. Its impact varies dramatically by operator, depending on whether they have invested in local processing capability:

- Huayou Cobalt (Arcadia): Insulated. Lithium sulphate plant commissioned October 2025; first export April 2026. The export ban does not cover lithium sulphate. Huayou's supply chain is buffered.
- Sinomine / 中矿资源 (Bikita): More exposed. Bikita's scale (approximately 250kt concentrate / 55kt LCE in 2023) makes it the most significant Zimbabwe supply source affected. The timeline for equivalent local processing capability is uncertain.
- Other small operators: Most exposed. Smaller concentrate-only operations face the full force of the export restriction with no local processing alternative.

The aggregate market effect is the removal of Zimbabwe's concentrate export volumes from global supply — estimated at approximately 7% of global spodumene supply based on 2025 total Zimbabwe exports of approximately 1.128 million tonnes. But the operator-specific effect is a competitive differentiation: operators with local processing capability gain relative market position.

3.4 Finding Four: The LCE Database Reveals the True Supply Concentration

When all confirmed operating production is converted to a common LCE basis, the concentration of reliable global lithium supply becomes apparent. Three assets — Greenbushes, Pilgangoora, and Salar de Atacama (SQM) — account for the majority of high-confidence supply:

Greenbushes (325kt LCE, FY25) + Pilgangoora (166kt LCE, FY25) + SQM Atacama (~170kt LCE estimated, 2025) = approximately 660kt LCE from three assets. Global lithium demand in 2025 was approximately 900,000–1,000,000 tonnes LCE. The remaining 240–340kt LCE must come from Wodgina, Cauchari-Olaroz, Jiajika, Jiangxi lepidolite, Arcadia, and all other sources combined.

This concentration has two implications. First, production disruptions at any of the top three assets — a grade decline at Greenbushes, a weather event at Pilgangoora, a governance dispute at Atacama — disproportionately affect global supply balances. Second, the diversity of the 'tail'

supply (Jiajika, Jiangxi lepidolite, Arcadia, Bikita) is exactly where price-sensitive and policy-sensitive supply exits and enters — making it the most important component for understanding price volatility even though it accounts for a minority of total volume.

Part IV-A — Supply Scenario Implications: Procurement Perspective

(This section is written from a procurement decision-making perspective. For trading strategy implications, see Part IV-B.)

4.1 Applying the Three Scenarios to Procurement Strategy

The three supply scenarios — Base, Tight, and Loose — are not forecasts. They are frameworks for testing procurement assumptions against specific observable triggers. A procurement strategy that performs reasonably well across all three scenarios is more robust than one optimised for a single price path.

Base Supply Case: Stable-to-Moderate Growth

In the base case, operating mines continue at guidance-level production, Arcadia's sulphate transition proceeds smoothly, and Jiangxi lepidolite stabilises at reduced but non-zero output. Global supply grows modestly — approximately 3–7% annually — driven primarily by Pilgangoora's P1000 expansion and Cauchari-Olaroz approaching nameplate.

Procurement implication: Normal cadence. Spot procurement of 30–40% of volume requirement is sustainable. Long-term offtake contracts at current prices (approximately USD 20,000–24,000/t) provide reasonable protection without overpaying if the Loose case materialises.

Tight Supply Case: Marginal Supply Exits

In the tight case, Jiangxi lepidolite contracts further (permit delays, environmental restrictions, or margin compression), Zimbabwe's concentrate ban is strictly enforced with limited quota relief, and Mt Cattlin does not restart. The marginal LCE supply reduction is estimated at 80–150kt LCE versus the base case — a meaningful deficit given that global demand growth of 10–15% per year implies incremental demand of approximately 100–150kt LCE annually.

Procurement implication: Increase long-term offtake exposure immediately. The tight case, if it materialises, will price in quickly once quarterly data confirms the supply contraction. Procurement teams that wait for price confirmation will pay a significant premium. Secure 60–70% of volume requirements through multi-year contracts with Australian hard-rock producers (Pilbara Minerals, Greenbushes).

Loose Supply Case: Optional Supply Re-enters

In the loose case, price recovery above USD 20,000/t for two or more consecutive quarters triggers Mt Cattlin restart decisions, Manono announces meaningful first production, and Jiangxi lepidolite operations accelerate. This case is the least likely in the 2026–2027 window — Mt Cattlin restart requires a minimum three to six months of sustained margin recovery, and Manono's legal situation makes near-term production genuinely uncertain — but it caps the upside.

Procurement implication: Reduce spot exposure and defer long-term commitments above USD 22,000/t. If CAAM data weakens simultaneously (NEV growth below 10% YoY), the loose supply case meeting a demand disappointment could produce a rapid price reversal.

Part IV-B — Supply Scenario Implications: Trading Strategy Perspective

(This section replaces Part IV-A for LinkedIn/public distribution. It is written from a trading strategy perspective rather than a procurement perspective.)

4.3 Using the Supply Framework for GFEX Trading Decisions

The three supply scenarios provide observable, mine-level signals that can be used to update GFEX trading positions systematically. The advantage of mine-level analysis is that the signals — quarterly production reports, permit renewal decisions, government policy announcements — are observable before they are reflected in price data.

The Supply Signal Hierarchy

Not all supply signals have equal informational content. The following hierarchy reflects both the timeliness and the market-moving significance of each signal:

Signal	Lead Time	Impact	Trading Implication
Mt Cattlin restart announcement (Rio Tinto)	3–6 months to production	HIGH (~29kt LCE)	Reduce GFEX long exposure; deferred calendar spread pressure
CATL permit renewal (or denial)	Immediate output effect	MED (~46kt LCE capacity)	Permit denial = bullish; renewal = bearish for marginal supply
Zimbabwe quota volume announcements	Monthly policy updates	MED (~55kt LCE from Bikita alone)	Tighter quota = GFEX long; looser = reduce exposure
Pilbara BMX auction price	Each auction ~quarterly	Indirect — refiner margins	BMX >USD 1,200/t = restart signal; <USD 800/t = cut signal within 6 weeks
Greenbushes quarterly guidance revision	Quarterly	HIGH (world's largest mine)	Downward revision = extend longs; upward = reduce
Manono first production confirmation	Unknown	LOW (initial output small)	Psychological ceiling; may cap rally even before tonnes arrive

Calendar Spread Strategy Based on Supply Scenarios

The supply scenario framework provides specific guidance for GFEX calendar spread positioning — a lower-risk way to express supply timing views than outright directional positions.

- Tight supply case: Long deferred contracts (2027–2028) vs. short near-term (2026). The investment drought impact is most acute in 2027–2028, when the cancelled supply pipeline becomes visible. Deferred contracts should strengthen relative to spot as this timeline becomes apparent.
- Loose supply case: Reverse the spread — long near-term, short deferred — if Mt Cattlin restart and Manono production announcements coincide. This reflects new supply

entering the market in the medium term while near-term physical demand remains strong.

- Base supply case: Flat calendar spread; limited spread opportunity. Focus on outright directional positioning based on demand signals (CAAM data, GFEX inventory).

The Cost Curve as a Position Sizing Tool

Mine-level cost data provides a principled basis for position sizing that is more robust than technical analysis alone. Specifically:

- Below USD 9,000/t: All hard rock operations except Greenbushes and Cauchari-Olaroz are producing below or near cash cost. Historical evidence (June 2025) confirms this is an unsustainable price level. Maximum long exposure is justified when prices approach this level.
- USD 9,000–14,000/t: Normal operations for low-cost mines; high-cost lepidolite begins exiting. Moderate long bias appropriate. Monitor CAAM data for demand confirmation.
- USD 14,000–20,000/t: Comfortable margin for all operating mines; Mt Cattlin restart becomes economically rational above USD 18,000/t. Reduce long exposure as USD 18,000/t approaches; watch for Rio Tinto announcements.
- Above USD 20,000/t: Sodium-ion substitution ceiling becomes binding for grid storage applications. Limit or reduce long exposure. Options strategy: buy puts at USD 18,000/t strike to cap downside if substitution accelerates.

Conclusion

Mine-level analysis changes the nature of the lithium price debate — from 'what will the aggregate supply-demand balance be?' to 'which specific assets will enter or exit the market, on what timeline, and with what consequences for the cost curve?' The latter question has specific, observable, falsifiable answers. The former does not.

Five Things This Analysis Adds to the Public Discourse

- The 30–40% supply pipeline reduction is quantifiable at the asset level: Mt Cattlin, Greenbushes guidance downgrade, CATL Jiangxi suspension, and Albemarle Wodgina deferrals are the specific, named components of a figure that is usually cited without attribution.
- Bikita and Arcadia are different assets owned by different companies. The market's conflation of these two — because both are Chinese-owned Zimbabwe projects — is a systematic analytical error with material implications for assessing Huayou Cobalt's exposure to Zimbabwe policy risk.
- Wodgina's 100% mine basis production requires quarterly reconstruction across the MinRes ownership change in October 2023. Using attributable production numbers directly produces a significant understatement of Wodgina's actual output, which in turn understates Australian supply and overstates the global deficit.
- The Salar de Atacama figures in public data are company-level sales, not mine-level production. They cannot be simply combined to produce a total Atacama output figure, and they include small volumes from other assets (Mt Holland for SQM in 2024).
- Jiangxi lepidolite is analytically most useful when treated as a price-sensitive regional supply pool rather than as a collection of individual mines. The uncertainty in its production data is a feature, not a bug — it reflects the price-responsive, fragmented nature of this supply source and makes it exactly the right variable to model in supply scenario analysis.

Monitoring Checklist

Weekly

- GFEX registered inventory (SMM / Mysteel): below 40,000t = watch; below 30,000t = acute tightness
- GFEX forward curve shape: contango → flat = transition; backwardation = acute tightness confirmed

Monthly

- CAAM NEV sales (~10th of each month): >20% YoY = bullish update; <10% YoY = bearish update
- GGII battery installation data: LFP share >75% = carbonate demand accelerating
- Chinese refinery operating rates: <70% = bullish; >85% = supply response under way

Quarterly / Event-Driven

- Rio Tinto Mt Cattlin: any restart announcement or underground study update
- Greenbushes quarterly production: IGO quarterly activities report — any guidance revision
- CATL Jiangxi: mining permit renewal status — immediate output impact on announcement
- Zimbabwe policy: quota volume allocations, export tax adjustments, and sulphate exemption scope
- Pilbara BMX auction results: spodumene price signal for Chinese refinery margin
- Huayou Annual / H1 Report: Arcadia production and sulphate output data

Xueting Xiang | xiangxuetingum@gmail.com | insightgatenews.com | May 2026

PhD Researcher, Quantitative Finance | Universiti Malaya | Published: Research in International Business and Finance (SSCI Q1 | ABDC-A)